

Casper DPM: Cascaded Perceptual Dynamic Projection Mapping onto Hands

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Abstract

We present a technique for dynamically projecting 3D content onto human hands with short perceived motion-to-photon latency. Computing the pose and shape of human hands accurately and quickly is a challenging task due to their articulated and deformable nature. We combine a slower 3D coarse estimation of the hand pose with high speed 2D correction steps which improve the alignment of the projection to the hands, increase the projected surface area, and reduce perceived latency. Since our approach leverages a full 3D reconstruction of the hands, any arbitrary texture or reasonably performant effect can be applied, which was not possible before. We conducted two user studies to assess the benefits of using our method. The results show subjects are less sensitive to latency artifacts and perform faster and with more ease a given associated task over the naïve approach of directly projecting rendered frames from the 3D pose estimation. We demonstrate several novel use cases and applications.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**.

Keywords

Projection Mapping, Dynamic Projection Mapping, Perceptive Rendering

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1 Introduction

Dynamic Projection Mapping (DPM) is a highly challenging task yet with numerous visually pleasing applications. Augmenting the appearance of moving objects can be both entertaining and informative. However, dynamic projector based augmentations requires high precision and even higher performance. Humans are sensitive to even slight misalignment during projection [Peng et al. 2023] — a trait very difficult to satisfy for moving objects, as they need to be tracked in 3D, processed to projection images, and passed to the projector, while they continue to move.

This is especially true for human hands, which are highly articulate, and tend to move rapidly and sporadically. For this reason, even though hands are our most effective manipulation tools as humans, DPM onto body parts techniques in the literature address mostly facial and full-body motion, but rarely projection onto human hands. Related methods that do address hands impose restrictions to relax the problem (e.g., using wearables [Sueishi and Ishikawa 2021], using markers or avoid the palm area [Peng and Watanabe 2021], restricting the type of augmentations possible [Miyashita et al. 2018], addressing hands as occluders to another object of interest [Halvorson et al. 2022], etc.).

In this demo we introduce *Casper DPM*— a short motion-to-photon projection mapping framework that enables wearable-free full 3D augmentations with low perceived latency. Our cascaded solution combines a coarse pose estimator in 3D with accurate and rapid screen space corrections based on a high speed camera. With it, we first correct low frequency artifacts, including calibration misalignment, shape mismatches, skinning and tracking errors. Still, after this step edges do not align well during performance due to shape mismatches and latency issues. Hence, to improve perceived

latency, we further exploit the attention given by our perception of motion to silhouette edges, and propose a last-minute correction to cover the whole observed surface area.

We have implemented a prototype for evaluations, and have developed different experimental and entertaining applications. Our solution is the first to tackle DPM onto wearable-free hands with full 3D capabilities.

We envision our system enabling simple and immersive DPM onto hands, allowing new applications such as gaming, physiotherapy, social performances, and creative artwork.

2 Method

In terms of hardware, we employ a typical projection-camera setup [Bermano et al. 2017; Punpongsanon et al. 2020], where a high-speed projector operates in the visible spectra, and a coaxial high-speed camera is filtered to observe only IR reflections. To track the hands' pose, we add an off-the-shelf 3D pose estimator (LMC, [Ultraleap 2024]), allowing for roughly 10ms latency pose estimations using a stereo pair IR cameras. Note that any 3D pose estimator with similar performance can be used. We position the projector and camera in a collocated setup, and the LMC closer to the scene from above. The elements are mounted onto a rig rotated at roughly 45 degrees for tabletop viewing. The camera is covered with a 850nm IR pass filter, and a hot mirror is placed between it and the projector.

In terms of software, we add two additional steps to the traditional track-render-project pipeline, to compensate for misalignments, and perceptually reduce temporal artifacts. We first add a 2D deformation step to every rendered frame. This compensates for low-frequency tracking errors. We then introduce *Perceptual Boundary Reduction*, a just-in-time effect filling gaps between projected and observed hands, which constitute one of the most distinct visual artifacts, thus reducing perceived latency.

3 Experimental Setup

The computer controlling the system has a 4 core Intel Xeon(R) W-2223 CPU @ 3.6GHz, and an Nvidia GeForce RTX 2080 Ti GPU, with Windows 10 OS. The entire logic is written in C++ using a custom OpenGL engine. The projector is a three color channel Dynaflex [Watanabe and Ishikawa 2019], set to 1024x768 resolution, able of roughly 1000 FPS throughput and 3ms latencies, and connected to the PC using an optic cable and a custom driver. The camera is a Basler acA720-520um computer vision camera, connected through a USB port, and filtered using a 850nm cutoff-frequency low pass filter. The hot-mirror reflects >90% of light in the range 750 - 1125nm, and transmits >85% of light in the range 425 - 675nm. We use version 1 of the LMC and the Ultraleap Gemini (5th generation) platform for tracking. The IR illumination consists of a square array of 49 diffuse 850nm LEDs packaged together, originally purposed for night photography.

4 Applications

Space Shooter. We modified an open sourced space shooter game [Baticle 2016] for the purpose of showing a simple gaming application with the system. Players can move the space ship to the left or right by tilting their palm around the elbow-to-wrist axis, and shoot alien ships in front of the space ship.

Hand Projection Performance. Inspired by *Shadowgraphy*, the art of performing a story or show using images made by hand shadows, we developed a new concept art form for projecting live content onto hands depending on their shape and form. We name this *Projography*. Here, the hand or hands form some intricate shape, and the system generates suitable adaptive content for that shape. For example, the hand imitates some features of an animal, and the system projects relevant content without any additional user input. This can be used both for performances, and for social gaming ("guessing the animal"). To do this live during the performance, we first employ the multimodal model GPT-4 [Achiam et al. 2023] to identify the animal that most resembles the current hand gesture by showing it a binary mask of the current hand silhouette. Then, ControlNet [Zhang et al. 2023] conditioned on Canny edge images is utilized with the same mask as a control signal, to generate an image of the identified animal. The generated image is baked onto the mesh using a projective texture and displayed using the real-time pipeline. This whole cycle takes between 2-8 seconds per animal and is launched on a separate thread, so the performer can animate the previously generated animal with no interference.

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